

WHAT IT IS

loaders.gl is an open source, framework-independent TypeScript/JavaScript suite of loaders and writers for visualization-oriented file formats, with emphasis on geospatial, 3D, tabular, image, tiled, and big data workflows.

WHO IT IS FOR

Developers building browser or Node.js visualization, analytics, geospatial, 3D, WebGL/deck.gl, backend, or cloud-service applications that need portable data loading into plain objects and typed arrays.

HOW TO RUN

- Repo development: Node >= 20.18.0, then yarn install.
- Build modules: yarn build or yarn build-modules.
- Quick tests: yarn test node; full repo tests: yarn test.
- Website build: cd website; yarn; yarn build.
- Use in an app: yarn add @loaders.gl/core plus a format module such as @loaders.gl/csv, then call load or parse.
- Runnable root app entrypoint: Not found in repo; root yarn start opens the loaders.gl docs site.

HOW IT WORKS

Workspace layout: Root package.json defines Yarn workspaces for modules/*, apps/*, dev-modules/*, examples, and website. The repo contains many @loaders.gl/* packages plus the @loaders.gl/tile-converter app.

Core API: modules/core/src/index.ts re-exports file fetching, loader selection, parse/load APIs, batch APIs, encoding APIs, data sources, iterator helpers, and worker/null loaders.

Data flow: load(url, loader) fetches string URLs or Blob/File-like input, sets a base URL, and delegates to parse. parse resolves data, selects a loader, normalizes options/context, and invokes a text parser, worker parser, or async parser.

Loader modules: Format modules such as csv, json, arrow, gltf, las, mvt, parquet, images, 3d-tiles, i3s, zip, and wms provide specific loaders/writers that plug into core.

Execution support: @loaders.gl/loader-utils supplies shared loader types/utilities; @loaders.gl/worker-utils supports worker execution; docs state the project avoids Node-specific APIs where possible for browser/Node portability.

Build/test flow: Root scripts use Ocular/Lerna for module builds, custom scripts for module tests and app commands, and a dedicated website build command.

WHAT IT DOES

- Loads and writes many file formats through composable npm modules.
- Covers table, geospatial, image, point cloud, mesh, scenegraph, and tiled data categories.
- Provides core APIs including load, parse, loadInBatches, parseInBatches, encode, and data sources.
- Auto-selects loaders from supplied candidates using extension, MIME type, or initial bytes.
- Supports browser and Node.js runtimes with portable ES module/ES2018 APIs.
- Supports worker parsing for many loaders and async-iterator batching for large or streaming data.
- Includes a tile-converter CLI/API app for conversion between formats such as 3D Tiles and I3S.